Yizhe Zhang

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Research Interest

My recent research focuses on pushing the LLM to gain more intuition and strong generalization, especially in the code domain, regarding: 1) code LLM/agent; 2) long-horizon planning; 3) RAG/reasoning with continuous tokens; 4) text diffusion models; 5) coding-based AI scientist.

Professional Experience

Staff Research Scientist, Apple, Cupertino, CA 05/2022 – Present

Manager: Navdeep Jaitly, Ronan Collobert

Using code domain as a testbed for planning and reasoning. Led development of:

- **Latent Reasoning: LaDiR, LaDiRL, CLaRa** – Continuous latent reasoning for LLMs
- **DiffuCoder** – Diffusion-based code generation (700 GitHub stars in one month)
- **SWE-Gym** – Code agents training framework with 1.8M+ dataset downloads per month

Research Scientist, Facebook AI, Menlo Park, CA 09/2021 – 05/2022

Manager: Yashar Mehdad

Worked on Retrieval-Augmented Generation (RAG) and Summarization systems.

Senior Researcher, Microsoft Research, Redmond, WA 02/2018 – 09/2021

Manager: Bill Dolan

Led development of generative dialogue models:

- **DialoGPT** – Open-sourced pretrained chat model with 1,912 citations, 2.4k GitHub stars

Education

Ph.D. in Computational Biology, Duke University, Durham, NC 08/2013 – 02/2018

Advisor: Prof. Lawrence Carin

Dissertation: Efficient and Scalable Markov Chain Monte Carlo Methods

Research Focus: VAE, GAN, MCMC for text generation

M.Sc. in Statistics, Duke University, Durham, NC 08/2015 – 02/2018

B.Sc. in Physics, Nanjing University, Nanjing, China 08/2007 – 06/2011

Professional Services

Area Chair: NeurIPS (since 2020), ICML (since 2022), ICLR (since 2023), ACL (2020-2021), EMNLP (2022), NAACL (2023), AAAI (2018-2021)

Action Editor: TMLR (since 2023), ARR (since 2023)

Organization Committee: ACL (2020)

Reviewer: NeurIPS, ICML, ICLR, ACL, EMNLP, NAACL, AISTATS, AAAI

Recent Talks

Towards Intuitive and Latent Thinking 04/2026

- Duke University Seminar, Invited speaker
- MSLD Workshop, UIUC, Invited speaker

Towards Understanding and Building Intuition for Language Model 10/2025

- University of Pennsylvania, Guest lecturer
- BAIR NLP workshop, UC Berkeley, Invited speaker
- University of Washington, NLP seminar, Invited speaker

Bidirectional Language Model 07/2025

- Apple Natural Language Understanding workshop, Invited speaker

Publications

Selected Preprints

[1]: Ruixiang Zhang, Richard He Bai, Huangjie Zheng, Navdeep Jaitly, Ronan Collobert, **Yizhe Zhang**. Embarrassingly Simple Self-Distillation Improves Code Generation. arXiv (2026)

[2]: Zidi Xiu, David Q. Sun, Kevin Cheng, Maitrik Patel, Josh Date, **Yizhe Zhang**, Jiarui Lu, Omar Attia, Raviteja Vemulapalli, Oncel Tuzel, Meng Cao, Samy Bengio. ASTRA-bench: Evaluating Tool-Use Agent Reasoning and Action Planning with Personal User Context. arXiv (2026)

[3]: Jie He, Richard He Bai, Sinead Williamson, Jeff Z. Pan, Navdeep Jaitly, **Yizhe Zhang**. CLaRa: Bridging Retrieval and Generation with Continuous Latent Reasoning. arXiv (2025) *[1k GitHub stars]*

[4]: **Yizhe Zhang**, Haoqiang Kang, Jie He, Navdeep Jaitly. LaDi-RL: Latent Diffusion with Reinforcement Learning for Test-Time Scaling. arXiv (2026) *[20.5 on AIME25 and 52.7 on LCB v6 for 8b model with 2x faster reasoning]*

[5]: **Yizhe Zhang**, Richard Bai, Zijin Gu, Ruixiang Zhang, Jiatao Gu, Emmanuel Abbe, Samy Bengio, Navdeep Jaitly. What makes the preferred thinking direction for LLM in Multi-choice Questions? arXiv (2025)

Peer-reviewed Conferences and Journals (* equally contributed)

[1]: Haoqiang Kang, **Yizhe Zhang**, Nikki Lijing Kuang, Nicklas Majamaki, Navdeep Jaitly, Yi-An Ma, Lianhui Qin. LaDiR: Latent Diffusion Enhances LLMs for Text Reasoning. ICLR (2026)

[2]: Huangjie Zheng, Shansan Gong, Ruixiang Zhang, Tianrong Chen, Jiatao Gu, Mingyuan Zhou, Navdeep Jaitly, **Yizhe Zhang**. Continuously Augmented Discrete Diffusion Model for Categorical Generative Modeling. ICLR (2026)

[3]: Amin Karimi Monse, Nikhil Bhendawade, Manuel Rafael Ciosici, Dominic Culver, **Yizhe Zhang**, Irina Belousova. FS-DFM: Fast and Accurate Long Text Generation with Few-Step Diffusion Language Models. ICLR (2026)

[4]: Shansan Gong, Ruixiang Zhang, Huangjie Zheng, Jiatao Gu, Navdeep Jaitly, Lingpeng Kong, **Yizhe Zhang**. DiffuCoder: Understanding and Improving Masked Diffusion Models for Code Generation. ICLR (2026) *[700 GitHub stars in one month, beats Gemini Diffusion and Qwen coder]*

[5]: Wei Liu, Ruochen Zhou, Yiyun Deng, Yuzhen Huang, Junteng Liu, Yuntian Deng, **Yizhe Zhang**, Junxian He. Learn to Reason Efficiently with Adaptive Length based Reward Shaping. ICLR (2026)

[6]: Deepro Choudhury, Sinead Williamson, Adam Goliński, Ning Miao, Freddie Bickford Smith, Michael Kirchhof, **Yizhe Zhang**, Tom Rainforth. BED-LLM: Intelligent Information Gathering with LLMs and Bayesian Experimental Design. ICLR (2026)

[**7**]: Ruixiang Zhang, Shuangfei Zhai, Jiatao Gu, **Yizhe Zhang**, Huangjie Zheng, Tianrong Chen, Miguel Angel Bautista, Josh Susskind, Navdeep Jaitly. Flexible Language Modeling in Continuous Space with Transformer-based Autoregressive Flows. NeurIPS (2025)

[**8**]: Jiayi Pan, Xingyao Wang, Graham Neubig, Navdeep Jaitly, Heng Ji, Alane Suhr, **Yizhe Zhang**. Training Software Engineering Agents and Verifiers with SWE-Gym. ICML (2025) *[1.8M+ dataset downloads per month]*

[**9**]: Ruixiang Zhang, Shuangfei Zhai, **Yizhe Zhang**, James Thornton, Zijing Ou, Joshua Susskind, Navdeep Jaitly. Target Concrete Score Matching: A Holistic Framework for Discrete Diffusion. ICML (2025)

[**10**]: Yifu Qiu, Varun Embar, **Yizhe Zhang**, Navdeep Jaitly, Shay B. Cohen, Benjamin Han. Eliciting In-context Retrieval and Reasoning for Long-context Large Language Models. ACL Findings (2025)

[**11**]: Shansan Gong, Shivam Agarwal, **Yizhe Zhang**, Jiacheng Ye, Lin Zheng, Mukai Li, Chenxin An, Peilin Zhao, Wei Bi, Jiawei Han, et al. Scaling Diffusion Language Models via Adaptation from Autoregressive Models. ICLR (2025)

[**12**]: Xingyao Wang, Boxuan Li, Yufan Song, Frank F. Xu, Xiangru Tang, Mingchen Zhuge, Jiayi Pan, Yueqi Song, Bowen Li, Jaskirat Singh, et al. All-hands: An Open Platform for AI Software Developers as Generalist Agents. ICLR (2024)

[**13**]: Jiatao Gu, Yuyang Wang, **Yizhe Zhang**, Qihang Zhang, Dinghuai Zhang, Navdeep Jaitly, Josh Susskind, Shuangfei Zhai. DART: Denoising Autoregressive Transformer for Scalable Text-to-Image Generation. ICLR (2024)

[**14**]: Guoli Yin, Haoping Bai, Shuang Ma, Feng Nan, Yanchao Sun, Zhaoyang Xu, Shen Ma, Jiarui Lu, Xiang Kong, Aonan Zhang, et al. MMAU: A Holistic Benchmark of Agent Capabilities Across Diverse Domains. NAACL Findings (2025)

[**15**]: Jiarui Lu, Thomas Holleis, **Yizhe Zhang**, Bernhard Aumayer, Feng Nan, Felix Bai, Shuang Ma, Shen Ma, Mengyu Li, Guoli Yin, et al. Toolsandbox: A Stateful, Conversational, Interactive Evaluation Benchmark for LLM Tool Use Capabilities. NAACL Findings (2025)

[**16**]: Jiatao Gu*, Ying Shen*, Shuangfei Zhai, **Yizhe Zhang**, Navdeep Jaitly, Joshua M. Susskind. Kaleido Diffusion: Improving Conditional Diffusion Models with Autoregressive Latent Modeling. NeurIPS (2024)

[**17**]: Yong Lin, Skyler Seto, Maartje Ter Hoeve, Katherine Metcalf, Barry-John Theobald, Xuan Wang, **Yizhe Zhang**, Chen Huang, Tong Zhang. On the Limited Generalization Capability of the Implicit Reward Model Induced by Direct Preference Optimization. EMNLP Findings (2024)

[**18**]: Zhuofeng Wu, He Bai, Aonan Zhang, Jiatao Gu, VG Vydiswaran, Navdeep Jaitly, **Yizhe Zhang**. Divide-or-Conquer? Which Part Should You Distill Your LLM? EMNLP Findings (2024)

[**19**]: **Yizhe Zhang***, He Bai*, Ruixiang Zhang*, Jiatao Gu, Shuangfei Zhai, Josh Susskind, Navdeep Jaitly. How Far Are We from Intelligent Visual Deductive Reasoning? COLM (2024)

[**20**]: **Yizhe Zhang**, Jiarui Lu, Navdeep Jaitly. The Entity-Deduction Arena: A playground for probing the conversational reasoning and planning capabilities of LLMs. ACL (2024)

[**21**]: Pratyush Maini, Skyler Seto, He Bai, David Grangier, **Yizhe Zhang**, Navdeep Jaitly. Rephrasing the web: A recipe for compute and data-efficient language modeling. ACL (2024)

[**22**]: Dinghuai Zhang, **Yizhe Zhang**, Jiatao Gu, Ruixiang Zhang, Josh Susskind, Navdeep Jaitly, Shuangfei Zhai. Improving GFlowNets for Text-to-Image Diffusion Alignment. ICML SPIGM workshop (2024)

[**23**]: Xingyao Wang, Yangyi Chen, Lifan Yuan, **Yizhe Zhang**, Yunzhu Li, Hao Peng, Heng Ji. Executable

code actions elicit better llm agents. ICML (2024)

[**24**]: Jiatao Gu, Shuangfei Zhai, **Yizhe Zhang**, Lingjie Liu, Joshua M Susskind. Boot: Data-free distillation of denoising diffusion models with bootstrapping. ICML (2024)

[**25**]: Jiatao Gu, Shuangfei Zhai, **Yizhe Zhang**, Joshua M Susskind, Navdeep Jaitly. Matryoshka diffusion models. ICLR (2023)

[**26**]: **Yizhe Zhang**, Jiatao Gu, Zhuofeng Wu, Shuangfei Zhai, Joshua Susskind, Navdeep Jaitly. PLANNER: generating diversified paragraph via latent language diffusion model. NeurIPS (2023)

[**27**]: Shuangfei Zhai, Tatiana Likhomanenko, Etai Littwin, Dan Busbridge, Jason Ramapuram, **Yizhe Zhang**, Jiatao Gu, Joshua M Susskind. Stabilizing transformer training by preventing attention entropy collapse. ICML (2023)

[**28**]: Felix Faltings, Michel Galley, Baolin Peng, Kianté Brantley, Weixin Cai, **Yizhe Zhang**, Jianfeng Gao, Bill Dolan. Interactive text generation. EMNLP (2023)

[**29**]: Jiatao Gu, Shuangfei Zhai, **Yizhe Zhang**, Miguel Angel Bautista, Josh Susskind. f-dm: A multi-stage diffusion model via progressive signal transformation. ICLR (2023)

[**30**]: Gyuwan Kim, Jinhyuk Lee, Barlas Oguz, Wenhan Xiong, **Yizhe Zhang**, Yashar Mehdad, William Yang Wang. Bridging the training-inference gap for dense phrase retrieval. EMNLP Findings (2022)

[**31**]: **Yizhe Zhang***, Deng Cai*. Linearizing transformer with key-value memory. EMNLP (2022)

[**32**]: Tianyu Liu, **Yizhe Zhang**, Chris Brockett, Yi Mao, Zhifang Sui, Weizhu Chen, Bill Dolan. A token-level reference-free hallucination detection benchmark for free-form text generation. ACL (2022)

[**33**]: **Yizhe Zhang**, Siqi Sun, Xiang Gao, Yuwei Fang, Chris Brockett, Michel Galley, Jianfeng Gao, Bill Dolan. Joint retrieval and generation training for grounded text generation. AAAI (2022)

[**34**]: Zhisong Zhang, **Yizhe Zhang**, Bill Dolan. Towards More Efficient Insertion Transformer with Fractional Positional Encoding. EACL (2021)

[**35**]: Woon Sang Cho, **Yizhe Zhang**, Sudha Rao, Asli Celikyilmaz, Chenyan Xiong, Jianfeng Gao, Mengdi Wang, William B Dolan. Contrastive Multi-document Question Generation. EACL (2021)

[**36**]: Jiachang Liu, Dinghan Shen, **Yizhe Zhang**, Bill Dolan, Lawrence Carin, Weizhu Chen. What Makes Good In-Context Examples for GPT-3? ACL DEELIO workshop (2021)

[**37**]: Zeqiu Wu, Michel Galley, Chris Brockett, **Yizhe Zhang**, Bill Dolan. Automatic document sketching: Generating drafts from analogous texts. ACL Findings (2021)

[**38**]: Jungo Kasai, Hao Peng, **Yizhe Zhang**, Dani Yogatama, Yi Mao, Weizhu Chen, Noah A Smith. Finetuning Pretrained Transformers into RNNs. EMNLP (2021, Oral Presentation)

[**39**]: Dianqi Li, **Yizhe Zhang**, Hao Peng, Liqun Chen, Chris Brockett, Ming-Ting Sun, Bill Dolan. Contextualized perturbation for textual adversarial attack. NAACL (2021)

[**40**]: Woon Sang Cho, **Yizhe Zhang**, Sudha Rao, Asli Celikyilmaz, Chenyan Xiong, Jianfeng Gao, Mengdi Wang, Bill Dolan. Unsupervised Common Question Generation from Multiple Documents using Reinforced Contrastive Coordinator. EACL (2021)

[**41**]: Zeqiu Wu, Michel Galley, Chris Brockett, **Yizhe Zhang**, Xiang Gao, Chris Quirk, Rik Koncel-Kedziorski, Jianfeng Gao, Hannaneh Hajishirzi, Mari Ostendorf, Bill Dolan. A Controllable Model of Grounded Response Generation. AAAI (2021)

[**42**]: Ramakanth Pasunuru, Asli Celikyilmaz, Michel Galley, Chenyan Xiong, **Yizhe Zhang**, Mohit Bansal, Jianfeng Gao. Data Augmentation for Abstractive Query-Focused Multi-Document Summarization. AAAI (2021)

[**43**]: **Yizhe Zhang**, Xiang Gao, Sungjin Lee, Chris Brockett, Michel Galley, Jianfeng Gao, Bill Dolan. Consistent Dialogue Generation with Self-supervised Feature Learning. SIGDIAL (2020)

[**44**]: **Yizhe Zhang**, Guoyin Wang, Chunyuan Li, Zhe Gan, Chris Brockett, Bill Dolan. POINTER:

Constrained Text Generation via Insertion-based Generative Pre-training. EMNLP (2020)

[**45**]: Xiang Gao, **Yizhe Zhang**, Michel Galley, Chris Brockett, Bill Dolan. Dialogue Response Ranking Training with Large-Scale Human Feedback Data. EMNLP (2020)

[**46**]: Chunyuan Li, Xiang Gao, Yuan Li, Xiujun Li, Baolin Peng, **Yizhe Zhang**, Jianfeng Gao. Optimus: Organizing Sentences via Pre-trained Modeling of a Latent Space. EMNLP (2020)

[**47**]: Jianqiao Li, Chunyuan Li, Guoyin Wang, Hao Fu, Yuhchen Lin, Liqun Chen, **Yizhe Zhang**, Lawrence Carin. Improving Text Generation with Student-Forcing Optimal Transport. EMNLP (2020)

[**48**]: Yu Cheng, Zhe Gan, **Yizhe Zhang**, Oussama Elachqar, Dianqi Li, Jingjing Liu. Contextual Text Style Transfer. Findings of EMNLP (2020)

[**49**]: Shuyang Dai, Yu Cheng, **Yizhe Zhang**, Zhe Gan, Jingjing Liu, Lawrence Carin. Contrastively Smoothed Class Alignment for Unsupervised Domain Adaptation. ACCV (2020)

[**50**]: Xinnuo Xu, **Yizhe Zhang**, Lars Liden, Sungjin Lee. Datasets and Benchmarks for Task-Oriented Log Dialogue Ranking Task. Interspeech (2020)

[**51**]: Siyang Yuan, Ke Bai, Liqun Chen, **Yizhe Zhang**, Chenyang Tao, Chunyuan Li, Guoyin Wang, Ricardo Henao, Lawrence Carin. Weakly supervised cross-domain alignment with optimal transport Task. BMVC (2020)

[**52**]: **Yizhe Zhang**, Siqi Sun, Michel Galley, Yen-Chun Chen, Chris Brockett, Xiang Gao, Jianfeng Gao, Jingjing Liu, Bill Dolan. DialoGPT: Large-Scale Generative Pre-training for Conversational Response Generation. demo track, ACL (2020) *[1,912 citations, 2.4k GitHub stars]*

[**53**]: Yichen Huang*, **Yizhe Zhang***, Oussama Elachqar, Yu Cheng. INSET: Sentence Infilling with Inter-sentential Generative Pre-training. ACL (2020)

[**54**]: Pengyu Cheng, Renqiang Min, Dinghan Shen, Christopher Malon, **Yizhe Zhang**, Yitong Li, Lawrence Carin. Improving Disentangled Text Representation Learning with Information Theoretical Guidance. ACL (2020)

[**55**]: Xinjie Fan, **Yizhe Zhang**, Zhendong Wang, Mingyuan Zhou. Adaptive Correlated Monte Carlo for Contextual Categorical Sequence Generation. ICLR (2020)

[**56**]: Yuan Li, Chunyuan Li, **Yizhe Zhang**, Xiujun Li, Guoqing Zheng, Lawrence Carin, Jianfeng Gao. Complementary Auxiliary Classifiers for Label-Conditional Text Generation. AAAI (2020)

[**57**]: Liqun Chen, Ke Bai, Chenyang Tao, **Yizhe Zhang**, Guoyin Wang, Wenlin Wang, Ricardo Henao, Lawrence Carin. Sequence Generation with Optimal-Transport-Enhanced Reinforcement Learning. AAAI (2020)

[**58**]: Xiang Gao, **Yizhe Zhang**, Sungjin Lee, Michel Galley, Chris Brockett, Jianfeng Gao, Bill Dolan. Structuring latent spaces for stylized response generation. EMNLP (2019)

[**59**]: Dianqi Li, **Yizhe Zhang**, Zhe Gan, Yu Cheng, Chris Brockett, Ming-Ting Sun, Bill Dolan. Domain Adaptive Text Style Transfer. EMNLP (2019)

[**60**]: Xinnuo Xu, **Yizhe Zhang**, Lars Liden, Sungjin Lee. Unsupervised Dialogue Spectrum Generation for Log Dialogue Ranking. SIGDIAL (2019), Best paper nomination

[**61**]: Liqun Chen, Guoyin Wang, Chenyang Tao, Dinghan Shen, **Yizhe Zhang**, Lawrence Carin. Improving Textual Network Embedding with Global Attention via Optimal Transport. ACL (2019)

[**62**]: Dinghan Shen, Asli Celikyilmaz, **Yizhe Zhang**, Liqun Chen, Xin Wang, Jianfeng Gao, Lawrence Carin. Towards Generating Long and Coherent Text with Multi-Level Latent Variable Models. ACL (2019)

[**63**]: Xiang Gao, Sungjin Lee, **Yizhe Zhang**, Chris Brockett, Michel Galley, Jianfeng Gao, Bill Dolan. Jointly Optimizing Diversity and Relevance in Neural Response Generation. NAACL (2019)

[**64**]: Liqun Chen, **Yizhe Zhang**, Ruiyi Zhang, Chenyang Tao, Zhe Gan, Haichao Zhang, Bai Li, Dinghan

Shen, Changyou Chen, Lawrence Carin. Improving Sequence-to-Sequence Learning via Optimal Transport. ICLR (2019)

[**65**]: **Yizhe Zhang**, Michel Galley, Jianfeng Gao, Zhe Gan, Xiujun Li, Chris Brockett, Bill Dolan. Generating Informative and Diverse Conversational Responses via Adversarial Information Maximization. NeurIPS (2018)

[**66**]: Liqun Chen, Shuyang Dai, Chenyang Tao, Dinghan Shen, Zhe Gan, Haichao Zhang, **Yizhe Zhang**, Lawrence Carin. Adversarial Text Generation via Feature-Mover's Distance. NeurIPS (2018)

[**67**]: Yunchen Pu, Shuyang Dai, **Yizhe Zhang**, Zhe Gan, Lawrence Carin. Multi-Domain Joint Distribution Learning with Generative Adversarial Nets. ICML (2018)

[**68**]: Dinghan Shen, Guoyin Wang, Wenlin Wang, Martin Renqiang Min, Qinliang Su, **Yizhe Zhang**, Chunyuan Li, Ricardo Henao, Lawrence Carin. On Simple Word-Embedding-Based Models and Associated Pooling Mechanisms. ACL (2018)

[**69**]: Guoyin Wang, Chunyuan Li, Wenlin Wang, **Yizhe Zhang**, Dinghan Shen, Xinyuan Zhang, Ricardo Henao, Lawrence Carin. Joint Embedding of Words and Labels for Text Classification. ACL (2018)

[**70**]: Dinghan Shen, **Yizhe Zhang**, Ricardo Henao, Qinliang Su, Lawrence Carin. Deconvolutional Latent-Variable Model for Text Sequence Matching. AAAI (2018)

[**71**]: Wenlin Wang, Piyush Rai, Yunchen Pu, Kai Fan, **Yizhe Zhang**, Ricardo Henao, Lawrence Carin. A Flexible Probabilistic Framework for Learning to Predict Unseen Classes. AAAI (2018)

[**72**]: Wenlin Wang, Yunchen Pu, Vinay Verma, Kai Fan, **Yizhe Zhang**, Changyou Chen, Piyush Rai, Lawrence Carin. Zero-shot Learning via Class-Conditioned Deep Generative Models. AAAI (2018)

[**73**]: **Yizhe Zhang**, Dinghan Shen, Guoyin Wang, Zhe Gan, Ricardo Henao, Lawrence Carin. Deconvolutional Paragraph Representation Learning. NIPS (2017)

[**74**]: Zhe Gan, Liqun Chen, Weiyao Wang, Yunchen Pu, **Yizhe Zhang**, Lawrence Carin. Triangle Generative Adversarial Networks. NIPS (2017)

[**75**]: **Yizhe Zhang**, Changyou Chen, Zhe Gan, Lawrence Carin. Stochastic Gradient Monomial Gamma Sampler. ICML (2017)

[**76**]: **Yizhe Zhang**, Zhe Gan, Zhi Chen, Lawrence Carin. Adversarial Feature Matching for Text Generation. ICML (2017)

[**77**]: **Yizhe Zhang**, Xiangyu Wang, Changyou Chen, Lawrence Carin. Towards Unifying Hamiltonian Monte Carlo and Slice Sampling. NIPS (2016)

[**78**]: Changyou Chen, Nan Ding, Chunyuan Li, **Yizhe Zhang**, Lawrence Carin. Distributed Bayesian Learning with Stochastic Gradient MCMC. NIPS (2016)

[**79**]: **Yizhe Zhang**, Ricardo Henao, Lawrence Carin. Dynamic Poisson Factor Analysis. ICDM (2016)

[**80**]: Kai Fan, **Yizhe Zhang**, Katherine Heller. Triply Stochastic Variational Inference for Non-linear Beta Process Factor Analysis. ICDM (2016)

[**81**]: **Yizhe Zhang**, Ricardo Henao, Jianling Zhong, Lawrence Carin, Alexander Hartemink. Learning a Hybrid Architecture for Sequence Regression and Annotation. AAAI (2016)

[**82**]: **Yizhe Zhang**, Ricardo Henao, Chunyuan Li, Lawrence Carin. Bayesian Dictionary Learning with Gaussian Processes and Sigmoid Belief Networks. IJCAI (2016)

[**83**]: **Yizhe Zhang**, Changyou Chen, Ricardo Henao, Lawrence Carin. Laplacian Hamiltonian Monte Carlo. ECML (2016)

[**84**]: **Yizhe Zhang**, Yupeng He, Chaochun Wei. MOST+: a Motif Finding Approach Combining Genomic Sequence and Heterogeneous Genome-wide Signatures. BMC Genomics (2015)

[**85**]: Yupeng He, **Yizhe Zhang**, Guangyong Zheng, Chaochun Wei. CRF-based Transcription Factor Binding Site Finding System. BMC Genomics (2012)

[86]: Jiemeng Liu, Haifeng Wang, Hongxing Yang, **Yizhe Zhang**, Jinfeng Wang, Fangqing Zhao, Ji Qi. Composition-based Classification of Short Metagenomic Sequences Elucidates the Landscapes of Taxonomic and Functional Enrichment of Microorganisms. Nucleic Acids Research (2012)

Workshop, Demo, and Miscellaneous

[87]: Mingyuan Zhou, Yi Gu, Huangjie Zheng, Liangchen Song, Guande He, **Yizhe Zhang**, Wenze Hu, Yinfei Yang. Score Distillation of Flow Matching Models. arXiv (2025)

[88]: Xiaogeng Liu, Zhiyuan Yu, **Yizhe Zhang**, Ning Zhang, Chaowei Xiao. Automatic and Universal Prompt Injection Attacks Against Large Language Models. arXiv (2024)

[89]: Lucie Charlotte Magister, Katherine Metcalf, **Yizhe Zhang**, Maartje Ter Hoeve. On the Way to LLM Personalization: Learning to Remember User Conversations. ACL L2M2 Workshop (2025)

[90]: **Yizhe Zhang**, Navdeep Jaitly. SAGE: Steering Dialog Generation with Future-Aware State-Action Augmentation. EMNLP Workshop on NLPerspectives (2025)

[91]: Tatiana Likhomanenko, Luke Carlson, Richard He Bai, Zijin Gu, Han Tran, Zakaria Aldeneh, **Yizhe Zhang**, Ruixiang Zhang, Huangjie Zheng, Navdeep Jaitly. ChipChat: Low-Latency Cascaded Conversational Agent in MLX. ASRU demo track (2025)

[92]: Shangshang Zheng, He Bai, **Yizhe Zhang**, Yi Su, Xiaochuan Niu, Navdeep Jaitly. KGLens: Towards Efficient and Effective Knowledge Probing of Large Language Models with Knowledge Graphs. ACL Towards Knowledgeable Language Models Workshop (2024)

[93]: Georgia Gabriela Sampaio, Ruixiang Zhang, Shuangfei Zhai, Jiatao Gu, Josh Susskind, Navdeep Jaitly, **Yizhe Zhang**. TypeScore: A Text Fidelity Metric for Text-to-Image Generative Models. arXiv (2024)

[94]: Ying Shen, **Yizhe Zhang**, Shuangfei Zhai, Lifu Huang, Joshua M. Susskind, Jiatao Gu. Many-to-Many Image Generation with Auto-Regressive Diffusion Models. ICML 2024 Workshop on Structured Probabilistic Inference Generative Modeling (2024)

[95]: Shikib Mehri, Jinho Choi, Luis Fernando D'Haro, Jan Deriu, Maxine Eskenazi, Milica Gasic, Kallirroi Georgila, Dilek Hakkani-Tur, Zekang Li, Verena Rieser, et al. Report from the NSF Future Directions Workshop on Automatic Evaluation of Dialog: Research Directions and Challenges. NSF AED Workshop (2022)

[96]: Vighnesh Leonardo Shiv, Chris Quirk, Anshuman Suri, Xiang Gao, Khuram Shahid, Nithya Govindarajan, **Yizhe Zhang**, Jianfeng Gao, Michel Galley, Chris Brockett. Microsoft Icecaps: An Open-Source Toolkit for Conversation Modeling. ACL System Demonstrations (2019)

[97]: Woon Sang Cho, Pengchuan Zhang, **Yizhe Zhang**, Xiujun Li, Michel Galley, Chris Brockett, Mengdi Wang, Jianfeng Gao. Towards Coherent and Cohesive Long-form Text Generation. Workshop on Narrative Understanding (2019)

Teaching

Duke: Advanced Machine Learning (STA571), Instructor: Katherine Heller

Duke: Probabilistic Machine Learning (CS571), Instructor: Cynthia Rudin

Honors and Awards

2023–Present: Stanford top 2% scientists

2018: NeurIPS top 5% reviewer award

2008–2011: Department Fellowship

2012: National Excellent Graduate Scholarship (top 1%)

Travel Awards: NIPS (2015, 2016), ICML (2017), ICDM (2016), IJCAI (2016), AAAI (2016)

Technical Proficiency

Programming: PyTorch, TensorFlow, C/C++, Python, Java, Lua, MATLAB, R